RAOH

Researchers

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Advancing knowledge.
Creating real-world impact.

Systematic Review and Meta-Analysis (SRMA) — Detailed Guide

What is SRMA?

A **Systematic Review and Meta-Analysis (SRMA)** is a structured scientific method used to collect, critically appraise, and statistically combine results from multiple studies addressing the same research question.

It has two parts:

- **Systematic Review (SR):** structured collection + critical evaluation of studies
- **Meta-Analysis (MA):** statistical pooling of results

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1. When is SRMA used?

SRMA is used when:

- Multiple studies exist on the same topic
- Results are inconsistent or conflicting
- A more precise estimate is needed (e.g., prevalence, risk, effect size)

Examples:

- Prevalence of depression in medical students
- Effect of sleep deprivation on cognition
- Drug efficacy comparisons

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2. Who does SRMA?

Typical team roles:

- **Principal Investigator (PI):** leads the project, defines question
- **Reviewers (2–3 researchers):** screen studies independently
- **Data extractors:** extract study data
- **Statistician:** performs meta-analysis
- **Supervisor/Expert:** ensures scientific validity

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3. Step-by-Step Process

STEP 1: Define Research Question (PICO)

Use **PICO framework**:

- **P (Population):** Who? (e.g., medical students)
- **I (Intervention/Exposure):** What factor? (e.g., sleep deprivation)
- **C (Comparison):** control or no exposure
- **O (Outcome):** what is measured (e.g., depression)

Example:
Does sleep deprivation increase depression among medical students?

STEP 2: Write Protocol

Before starting:

Include:

- Objective
- Inclusion/exclusion criteria
- Databases
- Keywords
- Analysis plan

Register protocol in:

- PROSPERO

STEP 3: Literature Search

Search databases:

- PubMed
- Scopus
- Web of Science
- Google Scholar (supplementary)

Use Boolean operators:

Example:
(“sleep deprivation” OR “poor sleep”) AND (“medical students”) AND (“depression”)

STEP 4: Remove Duplicates

Use tools like:

- EndNote
- Zotero
- Rayyan

Duplicates are removed before screening.

STEP 5: Screening Studies (VERY IMPORTANT)

Phase 1: Title & Abstract screening

Remove irrelevant studies.

Phase 2: Full-text screening

Check full eligibility.

Use PRISMA flow process.

Usually done by:

- 2 independent reviewers
- Disagreements resolved by third reviewer

STEP 6: Apply Inclusion & Exclusion Criteria

Inclusion examples:

- Original studies
- Human subjects
- Relevant outcomes
- English language (sometimes)

Exclusion examples:

- Reviews
- Case reports (depending on design)
- Animal studies
- Irrelevant outcomes

STEP 7: Data Extraction

Extract using Excel or Google Sheets:

Typical variables:

- Author

- Year
- Country
- Study design
- Sample size
- Outcome measures
- Effect size (OR, RR, mean difference)

STEP 8: Quality Assessment (Risk of Bias)

Evaluate study quality using tools like:

- Newcastle-Ottawa Scale (NOS) → observational studies
- Cochrane Risk of Bias Tool → RCTs
- JBI tools → cross-sectional studies

This determines study reliability.

STEP 9: Statistical Analysis (Meta-Analysis)

If data is suitable:

Effect measures:

- Odds Ratio (OR)
- Risk Ratio (RR)
- Mean Difference (MD)
- Standardized Mean Difference (SMD)

Models:

- Fixed-effect model (low heterogeneity)
- Random-effects model (high heterogeneity — most common)

Heterogeneity:

Measured by **I^2 statistic**

- 0–25% = low
- 25–50% = moderate
- 50% = high heterogeneity

STEP 10: Software Used

- RevMan (Cochrane)
- STATA
- R (meta, metafor packages)

- Comprehensive Meta-Analysis (CMA)

STEP 11: Publication of Results

Outputs include:

- Forest plot (main result)
- Funnel plot (publication bias)
- PRISMA flow diagram

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4. PRISMA Guidelines

PRISMA is the reporting standard for SRMA.

Includes:

- Search strategy transparency
- Study selection flow
- Reporting checklist

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5. Advantages of SRMA

- High level of evidence (top of evidence pyramid)
- Combines multiple studies
- More precise estimates
- Identifies research gaps

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6. Limitations

- Publication bias
- Heterogeneity between studies
- Poor quality of included studies affects results
- Time-consuming process

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7. Common Mistakes

- Weak or unclear research question
- Missing protocol registration
- Poor screening methodology

- No risk of bias assessment
- Mixing unrelated studies
- Ignoring heterogeneity

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8. Simple Workflow Summary

1. Define question (PICO)
2. Write protocol
3. Search databases
4. Remove duplicates
5. Screen studies
6. Apply criteria
7. Extract data
8. Assess quality
9. Perform meta-analysis
10. Interpret results
11. Write according to PRISMA

Golden Rule

A good SRMA is systematic, transparent, reproducible, and statistically justified.